MeanFlow-Adapt: Adapting Pretrained Diffusion and Flow Models into 1-Step Generators for New Domains

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Abstract

Adapting pretrained diffusion and flow models to new domains usually retains their costly multi-step sampling and relies on a separate distillation stage to produce a few-step target domain generator. We propose **MeanFlow-Adapt (MFA)**, a universal framework that jointly adapts pretrained source generators to a target domain and distills them into fast MeanFlow generators. MFA first recovers a shared instantaneous-velocity representation from heterogeneous source models, allowing a unified initialization across x-, ϵ-, v-, and u-parameterized generators. It then fine-tunes the resulting MeanFlow model with a domain-guided iMF-style objective, where source-anchored guidance improves target-domain alignment under limited target-domain data. This yields a single transfer recipe across diffusion, denoising, flow, and MeanFlow-family source models while preserving efficient few-step sampling in the target domain. We evaluate MFA on ImageNet-pretrained DiT, SiT, JiT, iMF, and pMF models adapted to diverse labeled target domains. Our study investigates whether MeanFlow can serve as a common adaptation target for heterogeneous pretrained generators and support efficient target-domain generation with a favorable speed–quality trade-off.

Contributions.

- We provide the first work that combines flow-distillation into adaptation for flow and diffusion models
- We introduce a **universal instantaneous velocity recovery** procedure that maps heterogeneous source generators—including x-, ϵ-, v-, and u-predictors—to a common MeanFlow initialization.
- We propose a **domain-guided fine-tuning** objective that uses source-anchored guidance within an iMF-style training formulation to improve target-domain alignment during adaptation.
- We present a comprehensive **empirical study across heterogeneous source families**, to evaluate whether a unified MeanFlow objective can provide an effective speed–quality trade-off on new domains.

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1 Related Work

Fast Generation Recent work has devoted substantial effort to reducing the high sampling cost of diffusion models. From caching to faster attention mechanisms and pyramidal structures, to methods that lower the NFE by removing cost of guidance, or lower the number of denoising steps. Lowering denoising steps has been done by more efficient solvers and also distillation approaches. These approaches aim to compress many-step samplers into one-step or few-step generators while retaining sample quality, through consistency training, distribution-matching, or adversarial distillation. These models, after distillation remain confined to their teacher or initial training domain and remain susceptible to domain shifts. Further, distillation remains extremely susceptible to the architecture and the loss space of the model.

Adaptive Generation large pretrained generative models can be adapted to a new domain using limited target-domain data. few-shot finetuning, personalization, and parameter-efficient adaptation strategies that seek to improve target-domain realism while preserving useful source-domain priors. typically focus on domain alignment and controllability, without explicitly addressing the inference cost of the adapted generator. Adaptation of large generators is ok, but distilled generators oversaturate their capacity to adapt and fine-tuning the distilled model using a naive diffusion regression loss negates the benefits of distillation. PSO tries to adapt distilled diffusion models.

From Slow Source Domain Generators to Fast Target Domain Generators the intersection of acceleration and adaptation, seeking to transform strong but slow pretrained source-domain generators into efficient target-domain models. Two stage designs and one-stage design. Recently Uni-dad. Heavy computation and being model dependent. Limitations in complexity, generality, and model-family dependence.

We see if we can universally take any initial diffusion or flow-matching model, distilled via mean-flow or non-distilled and transfer it to a target domain. This gives us a universal distillation and adaptation pipeline, crafting fast high quality and diverse models on target domains.

1.1 Loss space and Prediction Space

1.2 Guidance

2 Methodology

2.1 Background

2.1.1 Flow-matching

Flow Matching (FM) [1–3] models a time-dependent velocity field that transports samples from a simple prior distribution to the data distribution. A common choice is the linear interpolation path $z_t = (1-t)x + t\epsilon$ with data $x \sim p_{\text{data}}$ and Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$. The corresponding *conditional velocity* along this path is $v^* = dz_t/dt = \epsilon - x$. FM trains a neural network (NN) v_θ to predict v^* by minimizing a regression objective:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t,x,\epsilon}\left[\|v_\theta(z_t, t) - v^*\|^2\right]. \tag{1}$$

However, a given intermediate point z_t may arise from multiple pairs (x, ϵ). Therefore, the true regression target is the *marginal velocity* [1], defined as $v(z_t, t) \triangleq \mathbb{E}[\,v^*, z_t\,]$, averaged over all pairs (x, ϵ) that produce the same z_t at time t. When clear, we omit t and write $v(z_t)$ instead of $v(z_t, t)$.

2.1.2 MeanFlow and improved MeanFlow

While Flow Matching learns the *instantaneous velocity* field $v(z_t)$, MeanFlow (MF) [4] instead considers the average velocity accumulated between two time steps r and t. It defines the *mean velocity* field as:

$$u(z_t, r, t) \triangleq \frac{1}{t-r}\int_r^t v(z_\tau)\, d\tau. \tag{2}$$

We omit the explicit dependence on (r, t) and write $u(z_t)$. Directly supervising the integral above is intractable during training. To avoid this, MF derives an identity that links the *instantaneous velocity* to the *mean velocity* through differentiation with respect to t:

$$v(z_t) = u(z_t) + (t-r)\,\frac{d}{dt}u(z_t). \tag{3}$$

where $du(z_t)/dt$ is computed through a Jacobian–vector product (`JVP`) in practice [4]. This relation, referred to as the *MeanFlow identity* [4], allows the original velocity regression objective to be rewritten in terms of a u-predictor NN. Using this identity, improved MeanFlow (iMF) [5] defines a *compound function* V_θ:

$$V_\theta(z_t) \triangleq u_\theta(z_t) + (t-r)\texttt{JVP}^{\texttt{sg}}(u_\theta; v_\theta). \tag{4}$$

where `sg` denotes stop-gradient. To allow flexible classifier-free guidance (CFG) at sampling time, iMF regresses a *guided target velocity* v_g^*:

$$v_{\text{g}}^* = v^* + \left(1 - \frac{1}{w}\right)\big(v_\theta(z_t, c) - v_\theta(z_t, \varnothing)\big), \tag{5}$$

where c and $\varnothing$ denote the presence and absence of a condition, respectively, and w is the guidance scale. Then, the *mean velocity* predictor and *compound function* become conditioned on both c and w, yielding $u_\theta(\cdot, c, w)$ and $V_\theta(\cdot, c, w)$. The *instantaneous velocity* v_θ may either be recovered from the *mean velocity* via $v_\theta(z_t, t) = u_\theta(z_t, t, t)$ or predicted by a separate v-head [5]. The model is then trained with an FM-like regression objective:

$$\mathcal{L}_{\text{iMF}} = \mathbb{E}_{t,r,x,c,\epsilon}\left[\left\|V_\theta(z_t, c, w) - v_g^*\right\|^2\right]. \tag{6}$$

This formulation can be viewed as u-prediction with v-loss. After training, iMF supports few-step sampling by evaluating u_θ over a coarse sequence of time intervals. Specifically, one-step sampling is achieved via $x = \epsilon - u_\theta(\epsilon, 0, 1, c, w)$.

2.2 Problem Formulation

We address source-to-target adaptation of pretrained generative models under heterogeneous source parameterizations. A pretrained *source generator* $\mathcal{M}_{\text{src}}$ may be formulated as an x-, ϵ-, v-, or u-predictor. Given limited target-domain samples, our goal is to jointly adapt $\mathcal{M}_{\text{src}}$ to the target distribution and distill it into a few-step MeanFlow *target generator* u_ϕ. This enables high-quality target-aligned generation while avoiding the costly multi-step sampling of x-, ϵ-, and v-parameterized teachers. We call the proposed method MeanFlow-Adapt (MFA).

2.3 MeanFlow-Adapt

We propose **MeanFlow-Adapt (MFA)**, a universal framework for joint target-domain adaptation and flow distillation from heterogeneous source generators. MFA is built on two key components. First, it introduces a *universal instantaneous velocity recovery* procedure that maps different source parameterizations to a shared velocity representation and enables consistent MeanFlow initialization. Second, it performs *domain-guided fine-tuning*, where the initialized MeanFlow model is adapted to the target domain through an iMF-style training objective with an improved guidance mechanism for stronger target-domain alignment.

Universal instantaneous velocity recovery. Pretrained source generators may operate in different prediction spaces, such as x-, ϵ-, v-, or u-prediction, which prevents a single adaptation procedure from being applied directly. MFA resolves this by recovering an *instantaneous velocity* from the source model output in a parameterization-agnostic way. This recovered velocity is then used to initialize the target MeanFlow model u_ϕ while preserving the original source design. Table 1 summarizes how each source parameterization is mapped to an instantaneous velocity and used for MFA initialization.

disentangle loss and prediction space somewhere, and mention pMF.

Table 1: Universal MFA initialization from heterogeneous source model parameterizations. Regardless of the original source-model loss space, MFA recovers an instantaneous velocity from the source prediction space and uses it to initialize the target MeanFlow model.

	x-**pred**	ϵ-**pred**	v-**pred**	u-**pred**
Source model pred	$x_\theta(z_t)$	$\epsilon_\theta(z_t)$	$v_\theta(z_t)$	$u_\theta(z_t),\ v^*_\theta(z_t)$
Recovered instant velocity v_θ	$\dfrac{x_t - x_\theta(z_t)}{t}$	$\dfrac{\epsilon_\theta(z_t) - x_t}{1-t}$	$v_\theta(z_t)$	$v^*_\theta(z_t)$
Mean velocity initialization u_ϕ	$u_\phi \leftarrow v_\theta$			$u_\phi \leftarrow u_\theta,\ v_\phi \leftarrow v^*_\theta$

$*$ We preserve the *source generator* design and do not add new heads. We use an existing velocity head when available (e.g., for iMF [5]); otherwise, we recover the *instantaneous velocity* via $v(z_t, t) = u(z_t, t, t)$.

Domain-guided fine-tuning (`DogFit`). After initialization, MFA adapts the MeanFlow target generator to the target domain through a guided iMF-style objective. Starting from the initialized MeanFlow model, we construct the corresponding instantaneous velocity predictor and optimize it toward a guided target velocity. Unlike standard guidance formulations, our guidance mechanism is anchored to the source initialization, which provides a stronger inductive bias toward target-domain alignment during fine-tuning:

$$v^*_{\texttt{DogFit}} = v^* + \left(1 - \frac{1}{w}\right)\big(v_\phi(z_t, c) - v_\theta(z_t, \varnothing)\big), \tag{7}$$

where $v_\theta(z_t, \varnothing)$ is derived from the fixed source model and $v_\phi(z_t, c)$ is derived from the learning target model. This allows stronger target-domain alignment and higher quality generation since $v_\phi(z_t, \varnothing)$ would be poorly trained (due to limited target dataset) and would lead to out of target distribution generation. Our regression-loss then optimizes:

$$\mathcal{L}_{\text{MFA}} = \mathbb{E}_{t,r,x,c,\epsilon}\left[\left\|V_\phi(z_t, c, w) - v^*_{\texttt{DogFit}}\right\|^2\right]. \tag{8}$$

introduce guidance a bit later, use stop gradient on guidance offset...

Time-step guidance. Guide away from previous time-steps instead of same time-steps to allow less number of denoising steps used.

Table 2: FID and $\text{FD}_{\text{DINOv2}}$ comparison across labeled target domains for diffusion, denoising, and flow-based teacher initializations. We report quality and the number of function evaluations (NFE) at inference.

Initialization	Method	NFE	Quality						
			Metric	ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
DiT/XL-2 [6] (ϵ-loss, ϵ-pred)	FT	250×2	FID ↓	20.83	24.07	5.03	11.77	10.60	14.46
	FT + iMF	1		–	–	–	–	–	–
	FT + iMF	2		–	–	–	–	–	–
	MFA	1		–	–	–	–	–	–
	MFA	2		–	–	–	–	–	–
	FT	250×2	$\text{FD}_{\text{DINOv2}}$ ↓	360.77	529.85	428.21	610.31	378.11	461.45
	FT + iMF	1		–	–	–	–	–	–
	FT + iMF	2		–	–	–	–	–	–
	MFA	1		–	–	–	–	–	–
	MFA	2		–	–	–	–	–	–
SiT/XL-2 [7] (v-loss, v-pred)	FT	250×2	FID ↓	9.66	26.10	4.92	7.76	10.53	11.79
	FT + iMF	1		–	–	–	–	–	–
	FT + iMF	2		–	–	–	–	–	–
	MFA	1		–	–	–	–	–	–
	MFA	2		–	–	–	–	–	–
	FT	250×2	$\text{FD}_{\text{DINOv2}}$ ↓	271.92	519.54	405.07	522.47	283.29	400.46
	FT + iMF	1		–	–	–	–	–	–
	FT + iMF	2		–	–	–	–	–	–
	MFA	1		–	–	–	–	–	–
	MFA	2		–	–	–	–	–	–
JiT/XL-2 [8] (x-loss, x-pred)	FT	50×2	FID ↓	–	–	–	–	–	–
	FT + iMF	1		–	–	–	–	–	–
	FT + iMF	2		–	–	–	–	–	–
	MFA	1		–	–	–	–	–	–
	MFA	2		–	–	–	–	–	–
	FT	50×2	$\text{FD}_{\text{DINOv2}}$ ↓	–	–	–	–	–	–
	FT + iMF	1		–	–	–	–	–	–
	FT + iMF	2		–	–	–	–	–	–
	MFA	1		–	–	–	–	–	–
	MFA	2		–	–	–	–	–	–

3 Results and Discussion

3.1 Experimental Setup

As the source model, we use pretrained ImageNet generators including DiT/XL-2 [6], SiT/XL-2 [7], JiT/XL-2 [8], iMF/XL-2 [5], and pMF/XL-2 [9]. These models cover x-, ϵ-, v-, and u-prediction settings, allowing us to evaluate MFA under a wide range of adaptation recipes. We adapt each source model using MFA to a collection of labeled target domains: ArtBench, Caltech, CUB-Birds, Food, and Stanford-Cars. This setup is designed to span diverse semantic domains and domain-shift magnitudes, in the spirit of prior transfer-learning studies. We compare MFA against standard fine-tuning (FT) and, for non-MeanFlow teachers, a two-stage baseline denoted *FT + iMF*, where the source model is first adapted to the target domain and then converted into an iMF-style fast generator. This comparison isolates the benefit of performing adaptation and distillation jointly via our method. For MeanFlow-family teachers (iMF and pMF), we only compare MFA against FT, as the model is already distilled. For all methods, we initialize from the corresponding pretrained source checkpoint and fine-tune on target-domain samples. At inference, we report both one-step and two-step generation when applicable. We evaluate generation quality using FID and $\text{FD}_{\text{DINOv2}}$, computed separately on each target domain and averaged across domains. We additionally report Inception Score (IS) in the appendix. Unless otherwise stated, all models use 256×256 image resolution. More implementation details are provided in the appendix.

Table 3: FID and $\text{FD}_{\text{DINOv2}}$ comparison across labeled target domains for MeanFlow-family teacher initializations. We report quality and the number of function evaluations (NFE) at inference.

Initialization	Method	NFE	Quality						
			Metric	ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
iMF/XL-2 [5] (v-loss, u-pred)	FT	1	FID ↓	–	25.55	–	–	–	–
	FT	2		–	–	–	–	–	–
	MFA	1		–	–	–	–	–	–
	MFA	2		–	–	–	–	–	–
	FT	1	$\text{FD}_{\text{DINOv2}}$ ↓	–	–	–	–	–	–
	FT	2		–	–	–	–	–	–
	MFA	1		–	–	–	–	–	–
	MFA	2		–	–	–	–	–	–
pMF/XL-2 [9] (v-loss, x-pred)	FT	1	FID ↓	–	–	–	–	–	–
	FT	2		–	–	–	–	–	–
	MFA	1		–	–	–	–	–	–
	MFA	2		–	–	–	–	–	–
	FT	1	$\text{FD}_{\text{DINOv2}}$ ↓	–	–	–	–	–	–
	FT	2		–	–	–	–	–	–
	MFA	1		–	–	–	–	–	–
	MFA	2		–	–	–	–	–	–

3.2 Main Results

3.3 Ablations

Fine-tune with a few epochs of not applying guidance and after that applying guidance, as well as guidance interval.

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A Appendix / supplemental material

Table 4: Inception Score (IS) comparison across labeled target domains for diffusion, denoising, and flow-based teacher initializations. We report quality and the number of function evaluations (NFE) at inference.

Initialization	Method	NFE	IS					
			ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
DiT/XL-2 [6] (ϵ-loss, ϵ-pred)	FT	250×2	–	–	–	–	–	–
	FT + iMF	1	–	–	–	–	–	–
	FT + iMF	2	–	–	–	–	–	–
	MFA	1	–	–	–	–	–	–
	MFA	2	–	–	–	–	–	–
SiT/XL-2 [7] (v-loss, v-pred)	FT	250×2	–	–	–	–	–	–
	FT + iMF	1	–	–	–	–	–	–
	FT + iMF	2	–	–	–	–	–	–
	MFA	1	–	–	–	–	–	–
	MFA	2	–	–	–	–	–	–
JiT/XL-2 [8] (x-loss, x-pred)	FT	50×2	–	–	–	–	–	–
	FT + iMF	1	–	–	–	–	–	–
	FT + iMF	2	–	–	–	–	–	–
	MFA	1	–	–	–	–	–	–
	MFA	2	–	–	–	–	–	–

Table 5: Inception Score (IS) comparison across labeled target domains for MeanFlow-family teacher initializations. We report quality and the number of function evaluations (NFE) at inference.

Initialization	Method	NFE	IS					
			ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
iMF/XL-2 [5] (v-loss, u-pred)	FT	1	–	–	–	–	–	–
	FT	2	–	–	–	–	–	–
	MFA	1	–	–	–	–	–	–
	MFA	2	–	–	–	–	–	–
pMF/XL-2 [9] (v-loss, x-pred)	FT	1	–	–	–	–	–	–
	FT	2	–	–	–	–	–	–
	MFA	1	–	–	–	–	–	–
	MFA	2	–	–	–	–	–	–